

DiaCausal — screens v2

P06 export · 3 October 2026 · Group 28, FCRIT · prepared for Member C (build) and Member D (safety review)

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

What is in this file

1. Status of every screen in the chat (unchanged / fixed in P05 / new)
2. Consistency and content audit, run by `handoff/check_screens.py`
3. Design tokens
4. Every screen, desktop 1280 and phone 390, full length

Summary

24 live screens (01–03, 06–26) plus the two Recommended files, which are now generated from 01 and 17. 04 and 05 were retired in P05. No screen is byte-for-byte unchanged, because the header fix touched every one; 03, 09–16 changed only in the shell (header, tabs, strip, composer).

P06 changed no content. It turned the last colour and radius literals into tokens, coloured icons through tokens, stopped the focus ring from changing a control's corner radius, and fixed the phone options table, which P05 had squeezed into one narrow column.

All numbers on these screens are example values; brackets such as R-[ID] and ₹[MIN]–[MAX] are placeholders for real rule IDs and prices.

1 • Status of every screen

SCREEN	STATUS	WHAT CHANGED
01-signin Sign in	fixed in P05	One form: email, password and 6-digit code. CAPTCHA removed. “Research prototype” badge removed; full sentence under the header and in the footer.
02-signin-error Sign in — details not matched	fixed in P05	Same vague message, now also covers a wrong or expired code. All three fields marked invalid together, so none is singled out.
03-signin-locked Sign in — locked	fixed in P05 (shell only)	Copy unchanged. Header badge removed; intended-use line and footer added.
04-signin-step2-code Two-step code	retired in P05	The 6-digit code moved into 01. No v2 file.
05-signin-step2-code-error Two-step code — rejected	retired in P05	A wrong or expired code is now handled by 02's single message. No v2 file.
06-mfa-setup Set up authenticator	fixed in P05	Copy now says email, password and code. Header badge removed; footer added.
07-intended-use How to use DiaCausal	fixed in P05	Names the three options; adds “does not tell you which option to add” and “does not show doses”. Duplicate intended-use callout removed (line and footer carry it).
08-session-ending Session ending	fixed in P05	Modal unchanged. Background is now the Patient Details tab: tab bar, strip without dose, composer without microphone.
09-notice-identifier Identifier removed	fixed in P05 (shell only)	Notice text unchanged. Tabs, header, strip and composer updated.
10-notice-out-of-scope Out of scope	fixed in P05 (shell only)	Notice text unchanged. Tabs, header, strip and composer updated.
11-notice-emergency Emergency	fixed in P05 (shell only)	Notice text unchanged. Tabs, header, strip and composer updated.
12-notice-blocked Cannot answer as written	fixed in P05 (shell only)	Notice text unchanged. Tabs, header, strip and composer updated.

SCREEN	STATUS	WHAT CHANGED
13-panel-empty Patient details — empty	fixed in P05 (shell only)	Panel unchanged. Now the left column of the Patient Details tab; on a phone it stays open because nothing is filled in.
14-panel-filled Patient details — filled	fixed in P05 (shell only)	Panel unchanged. On a phone the summary strip is collapsed and opens into the panel.
15-panel-out-of-range Patient details — out of range	fixed in P05 (shell only)	Panel unchanged. Stays open on a phone so the error is visible.
16-panel-example-data Patient details — example data	fixed in P05 (shell only)	Panel unchanged. Strip collapsed on a phone.
17-answer-options-compared Three options compared	fixed in P05	Headline “Three options compared for this patient”. Leader named on HbA1c only, because its interval vs DPP-4i excludes Ø. Sulfonylurea removed by a rule shows “Not estimated”. Safety cards show rule ID and source. New Evidence level and “vs DPP-4i” columns. “What drives this estimate” with 95% CIs. GLP-1 and ADA removed. Ends “The clinician decides.”
18-evidence-drawer Evidence drawer	fixed in P05	Adds page and licence; S01 WHO 2018 metadata; opens as a side sheet over 17 (bottom sheet on a phone).
19-insufficient-evidence Insufficient evidence	fixed in P05	Calmer wording; reasons tied to the 95% interval; points to Investigate; ends “The clinician decides.”
20-loading-stages Working through it	fixed in P05	Six stages reordered to the pipeline: input checks, safety rules, causal estimate, retrieval, writing, output checks.
21-answer-no-clear-difference No clear difference	new	Both intervals vs DPP-4i include Ø, so no leader; trade-offs come first; sulfonylurea is “Check first”.
22-investigate Investigate	new	Search box, example searches, sources list, cited passages with source, version, section, page and licence.
23-investigate-insufficient Investigate — no passage	new	Insufficient-evidence state. No generated answer.
24-analysis Analysis	new	Versions A–D table, SHAP A vs C, LIME stability. Labelled synthetic benchmark, example values.

SCREEN	STATUS	WHAT CHANGED
25-guide Guide	new	Intro, problem statement, solution, workflow, cautions, limitations, advantages.
26-about About	new	Four members with email placeholders, the guide, B.Tech Computer Engineering, FCRIIT.
diacausal-login Recommended login	fixed in P05	Now generated from 01, so it is identical to it.
diacausal-chat Recommended chat	fixed in P05	Now generated from 17, so it is identical to it.
00-preview Preview gallery	new	Every screen at 1280 and 390 side by side. A viewing aid, not a screen.
Design canvas A, B, C Three looks	superseded	Exploration only. Still contain GLP-1, ADA, single-drug headlines and microphones. Do not build from them.
Design canvas R-Login, R-Chat, R-Phone Recommended (canvas)	superseded	Replaced by diacausal-login, diacausal-chat and 01–26. Same problems as the three looks.
screens/ 01–20 (P03–P04 copies) First 20 screens	superseded	Replaced by screens-v2. Still carry the header badge, microphone, dose prompt and dose in the strip.

2 · Audit

Current screens (screens-v2, 26 files)

CHECK	RESULT
Same <style> block on every screen	✓ 1 distinct hash across 26 files
Same :root tokens	✓ 1 distinct hash
Same font link (Atkinson Hyperlegible + Source Serif 4)	✓ 1 distinct link
No colour literal outside :root (CSS and markup)	✓ 0 found
No corner-radius literal (except 50% for dots)	✓ 0 found
Only token font families	✓ 0 found
No single-drug instruction	✓ 0 found
No GLP-1	✓ 0 found
No ADA	✓ 0 found
No dose prompt	✓ 0 found
No microphone	✓ 0 found
No header badge	✓ 0 found
No dose in the patient strip	✓ 0 found
Intended-use sentence at least twice (line + footer)	✓ all 26 files
Every answer card ends “The clinician decides.”	✓ 17, 18, 19, 21 pass

Superseded copies still in the folders

screens/ 01–20, the old diacausal-login and diacausal-chat, and the Design canvas artboards.
Files with at least one problem:

PROBLEM	FILES
Header badge	22
Microphone	22
Dose prompt	18
Single-drug instruction	9
GLP-1	8
Dose in the patient strip	10
ADA	4
Answer without “The clinician decides.”	5

Move these to an archive folder so nobody builds from them.

CHECK

RESULT

Every estimate cell and driver has a 95% CI

✓ 0 missing

3 · Design tokens

Copied from every screen's :root and saved as handoff/tokens.css. Spacing inside small components (badge, table-cell and unit-chip padding) still uses 3-10 px literals; layout spacing uses the --s1...--s7 scale.

--ground	 #F4F7F3
--surface	 #FFFFFF
--ink	 #12211A
--ink-muted	 #3C4A42
--ink-placeholder	 #6B776F
--pine	 #0E5049
--pine-dark	 #083B36
--pine-hairline	 #7FB8A8
--on-pine-muted	 #CFE3DC
--border-strong	 #9FB0A6
--border-soft	 #C8D4CB
--track	 #DCE6DF
--safe-ink	 #14562A
--safe-line	 #1B6B33
--safe-fill	 #DEF0E2
--check-ink	 #6E4400
--check-line	 #6E4400
--check-fill	 #FBEBCB

--s3	12px
--s4	16px
--s5	22px
--s6	24px
--s7	32px
--r-control	8px
--r-status	10px
--r-card	12px
--border-w	2px
--h-input	56px
--h-button	58px
--h-touch	52px
--pine-fill	 #E4EEEB
--on-danger	 #FFFFFF
--scrim	 rgba(18,33,26,0.55)
--border-w-strong	3px
--font-mono	ui-monospace,'SFMono-Regular',Menlo,Consolas,monospace

--danger-ink	 #8A1010
--danger-line	 #8A1010
--danger-fill	 #FADCDC
--font-body	'Atkinson Hyperlegible', Verdana, system-ui, sans-serif
--font-display	'Source Serif 4', Georgia, serif
--s1	4px
--s2	8px

--row-muted	 #EAEFEB
--panel-w	360px
--drawer-w	420px
--tabbar-h	64px
--on-pine	 #FFFFFF
--r-badge	6px
--r-pill	999px
--content-w	1040px

One form: email, password and 6-digit code. CAPTCHA removed. “Research prototype” badge removed; full sentence under the header and in the footer.

Desktop 1280 · part 1 of 1

DiaCausal

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

Sign in

Compares three second-line options for adults with type 2 diabetes already taking metformin.

Email

name@hospital.example

Password

.....

6-digit code

From your authenticator app. It changes every 30 seconds.

000000

Sign in

Trouble signing in?

First time? Request an account

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

Phone 390 · part 1 of 2

DiaCausal

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

Sign in

Compares three second-line options for adults with type 2 diabetes already taking metformin.

Email

name@hospital.example

Password

.....

6-digit code

From your authenticator app. It changes every 30 seconds.

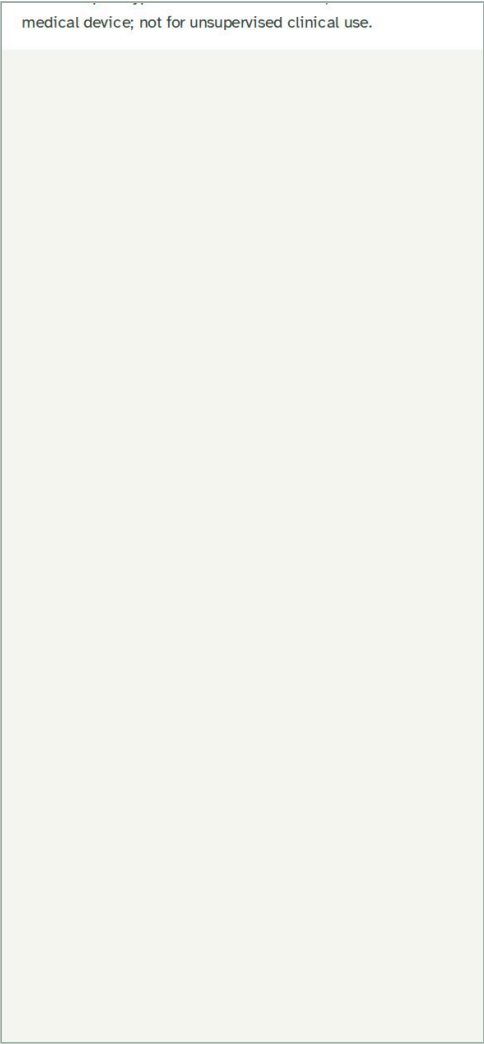
000000

Sign in

Trouble signing in?

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Research prototype for clinician evaluation; not a marketed



02-signin-error

· Sign in — details not matched

fixed in P05

Same vague message, now also covers a wrong or expired code. All three fields marked invalid together, so none is singled out.

Desktop 1280 · part 1 of 2

⌵

DiaCausal

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

Sign in

Compares three second-line options for adults with type 2 diabetes already taking metformin.

⊗ Those details did not match

Check your email, password and the code showing in your app now, then try again. [N] attempts left before sign-in locks.

Email

name@hospital.example

Password

.....

6-digit code

From your authenticator app. It changes every 30 seconds.

0 0 0 0 0 0

Sign in

Trouble signing in?

First time? Request an account

Phone 390 · part 1 of 2

⌵

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Email

name@hospital.example

Password

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6-digit code

From your authenticator app. It changes every 30 seconds.

0 0 0 0 0 0

Sign in

02-signin-error

· Sign in — details not matched

fixed in P05

Desktop 1280 · part 2 of 2

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

Phone 390 · part 2 of 2

Sign in

Trouble signing in?

First time? Request an account

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

Copy unchanged. Header badge removed; intended-use line and footer added.

Desktop 1280 · part 1 of 1

DiaCausal

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

Sign in

Signing in is locked

Too many attempts. Try again in [N] minutes or ask your administrator.

What to do now

Wait [N] minutes and sign in again. The lock clears on its own.

If you need access sooner, contact your administrator at [ADMIN CONTACT].

Sign in

Contact administrator

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

Phone 390 · part 1 of 1

DiaCausal

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

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Sign in

Contact administrator

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

06-mfa-setup · Set up authenticator

fixed in P05

Copy now says email, password and code. Header badge removed; footer added.

Desktop 1280 · part 1 of 2

DiaCausal

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

FIRST-TIME SETUP

Set up your authenticator app

You do this once. After today you sign in with your email, password and a 6-digit code.

1. Scan this code

[QR CODE IMAGE]

Cannot scan? Enter this key by hand

[ABCD][EFGH][IJKL][MNOP]

[QRST][UVWX]

Copy key

Keep this key private. Anyone with it can generate your codes.

What to do on your phone

1. Install **Google Authenticator** or **Microsoft Authenticator** from your phone's app store.

2. Open the app and choose **Add account**, then **Scan a QR code**.

3. Point your camera at the code on this page.

4. The app shows a 6-digit number that changes every 30 seconds.

Enter the 6-digit code to confirm

Confirm and finish setup

Phone 390 · part 1 of 2

DiaCausal

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

FIRST-TIME SETUP

Set up your authenticator app

You do this once. After today you sign in with your email, password and a 6-digit code.

1. Scan this code

[QR CODE IMAGE]

Cannot scan? Enter this key by hand

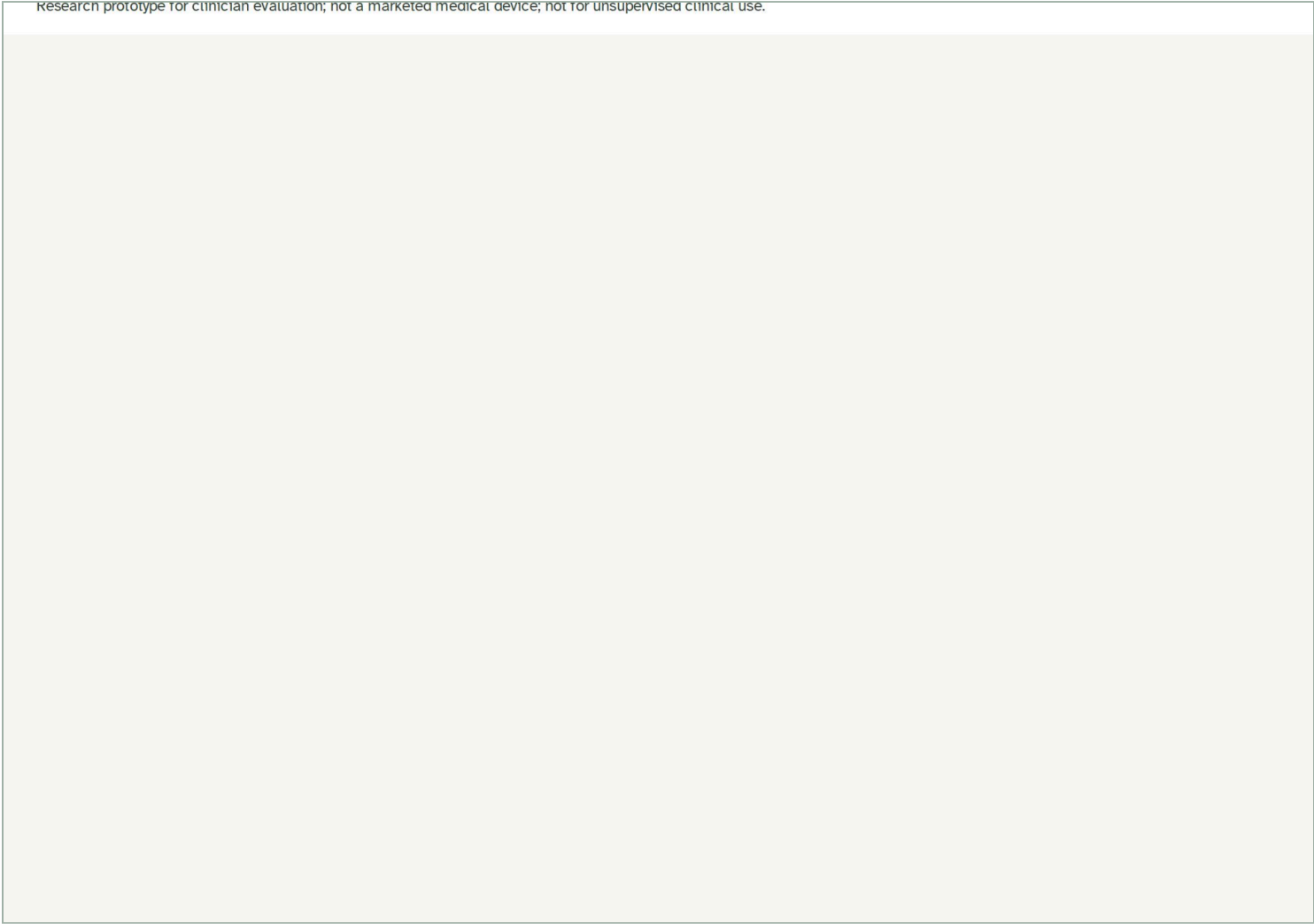
[ABCD][EFGH]

[IJKL][MNOP]

[QRST][UVWX]

Desktop 1280 · part 2 of 2

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.



Phone 390 · part 2 of 2

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000000

Confirm and finish setup

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

- How to use DiaCausal

fixed in P05

Names the three options; adds “does not tell you which option to add” and “does not show doses”. Duplicate intended-use callout removed (line and footer carry it).

Desktop 1280 · part 1 of 2

Phone 390 • part 1 of 2

⚡ DiaCausal

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

BEFORE YOU START

How to use DiaCausal

Read this once, and again each time the tool is updated.

What DiaCausal does

- ✓ Compares three second-line options — SGLT2 inhibitor, DPP-4 inhibitor and sulfonylurea — for an adult with type 2 diabetes already on metformin.
- ✓ Removes unsafe options with cited rules **before** estimating anything.
- ✓ Shows every estimate with a **95% interval**, never a bare number.
- ✓ Cites the source, version and section behind every clinical statement.
- ✓ Says “insufficient evidence” and stops when it cannot support an answer.

What DiaCausal does not do

- ❌ It does not tell you which option to add. You decide.
- ❌ It does not show doses, prescribe or place orders.
- ❌ It does not cover type 1 diabetes, pregnancy, under-18s or starting insulin.
- ❌ It does not accept patient identifiers. Do not type names, Aadhaar, phone, PAN or email.

DiaCausal

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

BEFORE YOU START

How to use DiaCausal

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- ✓ Cites the source, version and section behind every clinical statement.
- ✓ Says “insufficient evidence” and stops when it cannot support an answer.

What DiaCausal does not do

- ⊗ It does not tell you which option to add. You decide.

Desktop 1280 · part 2 of 2

⊗

It is not a marketed medical device, and its numbers come from synthetic data.

I understand

Continue

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

Phone 390 · part 2 of 2

⊗

It does not show doses, prescribe or place orders.

⊗

It does not cover type 1 diabetes, pregnancy, under-18s or starting insulin.

⊗

It does not accept patient identifiers. Do not type names, Aadhaar, phone, PAN or email.

⊗

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I understand

Continue

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

08-session-ending · Session ending fixed in P05

Modal unchanged. Background is now the Patient Details tab: tab bar, strip without dose, composer without microphone.

Desktop 1280 · part 1 of 2

DiaCausal

[USER NAME]

Sign out

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

Patient Details

Investigate

Analysis

Guide

About

Patient details

Example data

New patient

Clears these details and the conversation.

Age

18 or older.

58

years

Diabetes duration

6

years

HbA1c

8.4

%

eGFR

62

mL/min/1.73m²

BMI

31.2

kg/m²

Category: Obese (≥25)

Adult, 58 y · T2D 6 y · on metformin · HbA1c 8.4% · eGFR 62 · BMI 31.2

Example data

YOU ASKED

Compare the three options for this patient.

DIACAUSAL ANSWERED

[ANSWER CARD]

You'll be signed out in 2 minutes

This session has been idle. When it closes, the conversation on screen is cleared so the next person cannot read it.

Stay signed in

Sign out now

Phone 390 · part 1 of 1

DiaCausal

Sign out

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

PATIENT DETAILS

Adult, 58 y · T2D 6 y · on metformin · HbA1c 8.4% · eGFR 62 · BMI 31.2

Example data

YOU ASKED

DIACAUSAL ANSWERED

You'll be signed out in 2 minutes

This session has been idle. When it closes, the conversation on screen is cleared so the next person cannot read it.

Stay signed in

Sign out now

Ask about these three options for this patient

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

Patient Details

Investigate

Analysis

Guide

About

Desktop 1280 · part 2 of 2

Asian-Indian cut-offs: normal <23 · overweight 23–24.9 · obese ≥25

Established heart disease

✓ Yes

No

Chronic kidney disease

Yes

✓ No

Heart failure

Yes

✓ No

Past hypoglycaemia

✓ None

Mild

Severe

Budget for this drug

What the patient can spend each month.

₹ [AMOUNT]

₹ / month

Ask about these three options for this patient

↑

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

09-notice-identifier

· Identifier removed

fixed in P05 (shell only)

Notice text unchanged. Tabs, header, strip and composer updated.

Desktop 1280 · part 1 of 2

DiaCausal

[USER NAME]

Sign out

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

Patient Details

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Clears these details and the conversation.

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Category: Obese (≥25)

Adult, 58 y · T2D 6 y · on metformin · HbA1c 8.4% · eGFR 62 · BMI 31.2

Example data

YOU ASKED

Patient [NAME], Aadhaar [NUMBER], phone [NUMBER] — compare the options.

Identifier removed — not sent

Please remove patient identifiers (Aadhaar, phone, PAN, email) and ask again.

Describe the patient with the details panel instead.

Phone 390 · part 1 of 1

DiaCausal

Sign out

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

PATIENT DETAILS

▼

Adult, 58 y · T2D 6 y · on metformin · HbA1c 8.4% · eGFR 62 · BMI 31.2

Example data

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Desktop 1280 · part 2 of 2

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Established heart disease

✓ Yes

No

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Heart failure

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Past hypoglycaemia

✓ None

Mild

Severe

Budget for this drug

What the patient can spend each month.

₹ [AMOUNT]

₹ / month

Ask about these three options for this patient

↑

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

10-notice-out-of-scope · Out of scope

fixed in P05 (shell only)

Notice text unchanged. Tabs, header, strip and composer updated.

Desktop 1280 · part 1 of 2

DiaCausal

[USER NAME]

Sign out

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

Patient Details

Investigate

Analysis

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Example data

New patient

Clears these details and the conversation.

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Adult, 58 y · T2D 6 y · on metformin · HbA1c 8.4% · eGFR 62 · BMI 31.2

Example data

YOU ASKED

[QUESTION ABOUT TYPE 1 / PREGNANCY / UNDER 18 / STARTING INSULIN]

i

Out of scope

DiaCausal covers adults with type 2 diabetes already on metformin. This question is about [type 1 / pregnancy / under 18 / starting insulin].

Use your usual guideline, or refer as you normally would.

Phone 390 · part 1 of 1

DiaCausal

Sign out

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

PATIENT DETAILS

▼

Adult, 58 y · T2D 6 y · on metformin · HbA1c 8.4% · eGFR 62 · BMI 31.2

Example data

YOU ASKED

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Patient Details

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10-notice-out-of-scope

· Out of scope

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Desktop 1280 · part 2 of 2

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Established heart disease

✓ Yes

No

Chronic kidney disease

Yes

✓ No

Heart failure

Yes

✓ No

Past hypoglycaemia

✓ None

Mild

Severe

Budget for this drug

What the patient can spend each month.

₹ [AMOUNT]

₹ / month

Ask about these three options for this patient

↑

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

11-notice-emergency · Emergency fixed in P05 (shell only)

Notice text unchanged. Tabs, header, strip and composer updated.

Desktop 1280 · part 1 of 2

DiaCausal

[USER NAME]

Sign out

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

Patient DetailsInvestigateAnalysisGuideAbout

Patient details

Example data

New patient

Clears these details and the conversation.

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Adult, 58 y · T2D 6 y · on metformin · HbA1c 8.4% · eGFR 62 · BMI 31.2

Example data

YOU ASKED

[QUESTION DESCRIBING AN ACUTELY UNWELL PATIENT]

EMERGENCY

This may be an emergency. Follow your emergency protocol.

DiaCausal has not answered this question and will not suggest treatment for it.

Phone 390 · part 1 of 1

DiaCausal

Sign out

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

PATIENT DETAILS

Adult, 58 y · T2D 6 y · on metformin · HbA1c 8.4% · eGFR 62 · BMI 31.2

Example data

YOU ASKED

[QUESTION DESCRIBING AN ACUTELY UNWELL PATIENT]

EMERGENCY

This may be an emergency. Follow your emergency protocol.

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Mild

Severe

Budget for this drug

What the patient can spend each month.

₹ [AMOUNT]

₹ / month

Ask about these three options for this patient

↑

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

12-notice-blocked

· Cannot answer as written

fixed in P05 (shell only)

Notice text unchanged. Tabs, header, strip and composer updated.

Desktop 1280 · part 1 of 2

DiaCausal

[USER NAME]

Sign out

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

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%

eGFR

62

mL/min/1.73m²

BMI

31.2

kg/m²

Category: Obese (≥25)

Adult, 58 y · T2D 6 y · on metformin · HbA1c 8.4% · eGFR 62 · BMI 31.2

Example data

YOU ASKED

[MESSAGE THE FILTER DID NOT ACCEPT]

Cannot answer as written

Please rephrase your question.

Phone 390 · part 1 of 1

DiaCausal

Sign out

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

PATIENT DETAILS

Example data

YOU ASKED

[MESSAGE THE FILTER DID NOT ACCEPT]

Cannot answer as written

Please rephrase your question.

Ask about these three options for this patient

Patient DetailsInvestigateAnalysisGuideAbout

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

Desktop 1280 · part 2 of 2

Asian-Indian cut-offs: normal <23 · overweight 23–24.9 · obese ≥25

Established heart disease

✓ Yes

No

Chronic kidney disease

Yes

✓ No

Heart failure

Yes

✓ No

Past hypoglycaemia

✓ None

Mild

Severe

Budget for this drug

What the patient can spend each month.

₹ [AMOUNT]

₹ / month

Ask about these three options for this patient

↑

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

13-panel-empty · Patient details — empty fixed in P05 (shell only)

Panel unchanged. Now the left column of the Patient Details tab; on a phone it stays open because nothing is filled in.

Desktop 1280 · part 1 of 2

DiaCausal

[USER NAME]

Sign out

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

Patient Details

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About

Patient details

Not filled in

New patient

Clears these details and the conversation.

Age

18 or older.

years

Diabetes duration

years

HbA1c

0.0

%

eGFR

mL/min/1.73m²

BMI

0.0

kg/m²

Category: —

No patient details yet

Not filled in

Fill in the patient details to start

DiaCausal compares three second-line options for an adult with type 2 diabetes already on metformin. Nothing is compared until the details are in.

Do not type names, Aadhaar, phone, PAN or email.

Phone 390 · part 1 of 3

DiaCausal

Sign out

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

PATIENT DETAILS

No patient details yet

Not filled in

New patient

Clears these details and the conversation.

Age

18 or older.

years

Diabetes duration

years

HbA1c

0.0

%

eGFR

mL/min/1.73m²

BMI

0.0

kg/m²

Desktop 1280 · part 2 of 2

Asian-Indian cut-offs: normal <23 · overweight 23–24.9 · obese ≥25

Established heart disease

Yes

No

Chronic kidney disease

Yes

No

Heart failure

Yes

No

Past hypoglycaemia

None

Mild

Severe

Budget for this drug

What the patient can spend each month.

₹ [AMOUNT]

₹ / month

Ask about these three options for this patient

↑

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

Phone 390 · part 2 of 3

Category: —

Asian-Indian cut-offs: normal <23 · overweight 23–24.9 · obese ≥25

Established heart disease

Yes

No

Chronic kidney disease

Yes

No

Heart failure

Yes

No

Past hypoglycaemia

None

Mild

Severe

Budget for this drug

What the patient can spend each month.

₹ [AMOUNT]

₹ / month

ⓘ

Fill in the patient details to start

DiaCausal compares three second-line options for an adult with type 2 diabetes already on metformin. Nothing is compared until the details are in.

Do not type names. Aadhaar, phone, PAN

or email.

Ask about these three options for this patient

↑

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

Patient Details

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14-panel-filled

· Patient details — filled

fixed in P05 (shell only)

Panel unchanged. On a phone the summary strip is collapsed and opens into the panel.

Desktop 1280 · part 1 of 2

DiaCausal

[USER NAME] [Sign out](#)

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

Patient Details

Investigate

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Guide

About

Patient details

Entered by you

New patient

Clears these details and the conversation.

Age

18 or older.

58

years

Diabetes duration

6

years

HbA1c

8.4

%

eGFR

62

mL/min/1.73m²

BMI

31.2

kg/m²

Category: Obese (≥25)

Adult, 58 y · T2D 6 y · on metformin · HbA1c 8.4% · eGFR 62 · BMI 31.2

Entered by you

Details received

Compare the three options for this patient, or ask about them below.

Compare the three options

Phone 390 · part 1 of 1

DiaCausal

[Sign out](#)

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

PATIENT DETAILS

▼

Adult, 58 y · T2D 6 y · on metformin · HbA1c 8.4% · eGFR 62 · BMI 31.2

Entered by you

Details received

Compare the three options for this patient, or ask about them below.

Compare the three options

Ask about these three options for this patient

↑

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

Patient Details

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About

Desktop 1280 · part 2 of 2

Asian-Indian cut-offs: normal <23 · overweight 23–24.9 · obese ≥25

Established heart disease

✓ Yes

No

Chronic kidney disease

Yes

✓ No

Heart failure

Yes

✓ No

Past hypoglycaemia

✓ None

Mild

Severe

Budget for this drug

What the patient can spend each month.

₹ [AMOUNT]

₹ / month

Ask about these three options for this patient

↑

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

15-panel-out-of-range · Patient details — out of range fixed in P05 (shell only)

Panel unchanged. Stays open on a phone so the error is visible.

Desktop 1280 · part 1 of 2

DiaCausal

[USER NAME]

Sign out

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

Patient DetailsInvestigateAnalysisGuideAbout

Patient details

⚠ Check a value

New patient

Clears these details and the conversation.

Age

18 or older.

58

years

Diabetes duration

6

years

HbA1c

45

%

⚠ HbA1c 45%? Check the value — expected 4–20%.

eGFR

62

mL/min/1.73m²

BMI

One value needs checking

⚠ Check a value

⚠ One value needs checking

Options are not compared while a value is outside the expected range. Correct the highlighted field, then try again.

Phone 390 · part 1 of 3

DiaCausal

Sign out

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

PATIENT DETAILS

One value needs checking

⚠ Check a value

New patient

Clears these details and the conversation.

Age

18 or older.

58

years

Diabetes duration

6

years

HbA1c

45

%

⚠ HbA1c 45%? Check the value — expected 4–20%.

eGFR

62

mL/min/1.73m²

BMI

Desktop 1280 · part 2 of 2

31.2

kg/m²

Category: Obese (≥25)

Asian-Indian cut-offs: normal <23 · overweight 23–24.9 · obese ≥25

Established heart disease

✓ Yes

No

Chronic kidney disease

Yes

✓ No

Heart failure

Yes

✓ No

Past hypoglycaemia

✓ None

Mild

Severe

Budget for this drug

What the patient can spend each month.

₹ [AMOUNT]

₹ / month

Ask about these three options for this patient

↑

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

Phone 390 · part 2 of 3

31.2

kg/m²

Category: Obese (≥25)

Asian-Indian cut-offs: normal <23 · overweight 23–24.9 · obese ≥25

Established heart disease

✓ Yes

No

Chronic kidney disease

Yes

✓ No

Heart failure

Yes

✓ No

Past hypoglycaemia

✓ None

Mild

Severe

Budget for this drug

What the patient can spend each month.

₹ [AMOUNT]

₹ / month

⚠ One value needs checking

Options are not compared while a value is outside the expected range. Correct the highlighted field, then try again.

Ask about these three options for this patient

↑

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

Patient Details

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16-panel-example-data

· Patient details — example data

fixed in P05 (shell only)

Panel unchanged. Strip collapsed on a phone.

Desktop 1280 · part 1 of 2

DiaCausal

[USER NAME]

Sign out

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

Patient Details

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Patient details

Example data

New patient

Clears these details and the conversation.

Age

18 or older.

58

years

Diabetes duration

6

years

HbA1c

8.4

%

eGFR

62

mL/min/1.73m²

BMI

31.2

kg/m²

Category: Obese (≥25)

Adult, 58 y · T2D 6 y · on metformin · HbA1c 8.4% · eGFR 62 · BMI 31.2

Example data

Details received

Compare the three options for this patient, or ask about them below.

Compare the three options

Phone 390 · part 1 of 1

DiaCausal

Sign out

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

PATIENT DETAILS

▼

Adult, 58 y · T2D 6 y · on metformin · HbA1c 8.4% · eGFR 62 · BMI 31.2

Example data

Details received

Compare the three options for this patient, or ask about them below.

Compare the three options

Ask about these three options for this patient

↑

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

Patient Details

Investigate

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About

Desktop 1280 · part 2 of 2

Asian-Indian cut-offs: normal <23 · overweight 23–24.9 · obese ≥25

Established heart disease

✓ Yes

No

Chronic kidney disease

Yes

✓ No

Heart failure

Yes

✓ No

Past hypoglycaemia

✓ None

Mild

Severe

Budget for this drug

What the patient can spend each month.

₹ [AMOUNT]

₹ / month

Ask about these three options for this patient

↑

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

17-answer-options-compared

fixed in P05

Headline “Three options compared for this patient”. Leader named on HbA1c only, because its interval vs DPP-4i excludes 0. Sulfonylurea removed by a rule shows “Not estimated”. Safety cards show rule ID and source. New Evidence level and “vs DPP-4i” columns. “What drives this estimate” with 95% CIs. GLP-1 and ADA removed. Ends “The clinician decides.”

Desktop 1280 · part 1 of 3

[USER NAME] [Sign out](#)

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

[Patient Details](#)
[Investigate](#)
[Analysis](#)
[Guide](#)
[About](#)

Patient details

Example data

New patient

Clears these details and the conversation.

Age

18 or older.

58 years

Diabetes duration

6 years

HbA1c

8.4 %

eGFR

62 mL/min/1.73m²

BMI

31.2 kg/m²

Category: Obese (≥ 25)

Adult, 58 y · T2D 6 y · on metformin · HbA1c 8.4% · eGFR 62 · BMI 31.2 · past hypoglycaemia: severe

Example data

YOU ASKED

Compare the three options for this patient.

DIACAUSAL ANSWERED

Example values

Three options compared for this patient

Safety rules ran first. One of the three options was removed before any estimate was made.

Safety checks

SGLT2 inhibitor	DPP-4 inhibitor	Sulfonylurea
Safe to consider No exclusion rule matched. eGFR 62 is above this rule's kidney threshold.	Safe to consider No exclusion rule matched for this record.	Do not use Removed before estimation. It is not compared with the others.
Rule R-[ID] · checked on eGFR, type 1 diabetes, past ketoacidosis Source [SOURCE], [VERSION], § [SECTION] ③	Rule R-[ID] · checked on past pancreatitis, heart failure Source [SOURCE], [VERSION], § [SECTION] ③	Rule R-[ID] · triggered by past hypoglycaemia: severe Source [SOURCE], [VERSION], § [SECTION] ③

Phone 390 • part 1 of 6



DiCausal

[Sign out](#)

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

PATIENT DETAILS

Adult, 58 y · T2D 6 y · on metformin · HbA1c 8.4% · eGFR 62 · BMI 31.2 · past hypoglycaemia: severe

 Example data

YOU ASKED

Compare the three options for this patient.

DIACAUSAL ANSWERED

 Example values

Three options compared for this patient



Safety rules ran first. One of the three options was removed before any estimate was made.

Safety checks

SGLT2 inhibitor

 **Safe to consider**

No exclusion rule matched. eGFR 62 is above this rule's kidney threshold.

Rule R-[ID] · checked on eGFR, type 1 diabetes, past ketoacidosis

Desktop 1280 · part 2 of 3

Asian-Indian cut-offs: normal <23 · overweight 23–24.9 · obese ≥25

Established heart disease

✓ Yes

No

Chronic kidney disease

Yes

✓ No

Heart failure

Yes

✓ No

Past hypoglycaemia

None

Mild

✓ Severe

Budget for this drug

What the patient can spend each month.

₹ [AMOUNT]

₹ / month

HbA1c at 6 months

↑↓ **SGLT2 inhibitor shows more HbA1c lowering than DPP-4 inhibitor for this patient:**
difference −0.21 % (95% CI −0.36 to −0.06). The interval excludes zero. This is a difference on HbA1c only. ①

OPTION	HBA1C CHANGE, 6 MONTHS dot = estimate · bar = 95% CI · axis −1.6 to 0 %	DIFFERENCE VS DPP-4I	HYPO-GLYCAEMIA	WEIGHT, 6 MONTHS	COST	EVIDENCE LEVEL
SGLT2 inhibitor added to metformin ✓ Safe to consider	−1.05 % 95% CI −1.22 to −0.88 	−0.21 % 95% CI −0.36 to −0.06 ✓ Interval excludes 0	●○○ Low	−2.1 kg 95% CI −2.6 to −1.6	₹[MIN]–[MAX] per month	Moderate
↑↓ More HbA1c lowering than DPP-4i						
DPP-4 inhibitor added to metformin ✓ Safe to consider	−0.84 % 95% CI −0.99 to −0.69 	Comparator DPP-4 inhibitor is the reference	●○○ Low	−0.2 kg 95% CI −0.6 to +0.2	₹[MIN]–[MAX] per month	Moderate
Sulfonylurea added to metformin ⊗ Do not use	Not estimated Removed by rule R-[ID] before estimation	—	—	—	₹[MIN]–[MAX] per month	Not assessed

Evidence level — Moderate: a narrow interval and enough similar patients. Low: a wide interval or few similar patients. Insufficient: no estimate is shown. It is never “High”, because every estimate comes from a synthetic cohort.

Trade-offs

- **Weight:** SGLT2 inhibitor −2.1 kg (95% CI −2.6 to −1.6); DPP-4 inhibitor −0.2 kg (95% CI −0.6 to +0.2). ①
- **Hypoglycaemia:** both remaining options are in the low-risk category. ②
- **Cost:** compare each option against the patient's budget of ₹[AMOUNT] per month.

Phone 390 · part 2 of 6

Source [SOURCE], [VERSION], § [SECTION] ③

DPP-4 inhibitor

✓ **Safe to consider**

No exclusion rule matched for this record.

Rule R-[ID] · checked on past pancreatitis, heart failure

Source [SOURCE], [VERSION], § [SECTION] ③

Sulfonylurea

⊗ **Do not use**

Removed before estimation. It is not compared with the others.

Rule R-[ID] · triggered by past hypoglycaemia: severe

Source [SOURCE], [VERSION], § [SECTION] ③

HbA1c at 6 months

↑↓ **SGLT2 inhibitor shows more HbA1c lowering than DPP-4 inhibitor for this patient:**
difference −0.21 % (95% CI −0.36 to −0.06). The interval excludes zero. This is a difference on HbA1c only. ①

17-answer-options-compared · Three options compared

fixed in P05

Desktop 1280 · part 3 of 3

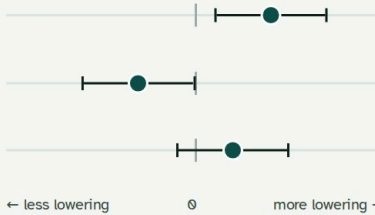
What drives this estimate

Difference between SGLT2 inhibitor and DPP-4 inhibitor for this patient. Top 3 drivers, in HbA1c points.

Higher starting HbA1c (this patient 8.4%): 0.04 more lowering (95% CI 0.01 to 0.07)

eGFR 62 mL/min/1.73m²: 0.03 less lowering (95% CI 0.00 to 0.06)

BMI 31.2 kg/m²: 0.02 more lowering (95% CI -0.01 to 0.05)



These show how the estimate changes across patients like this one. They are not causes.

Sources cited

- 1 DiaCausal causal engine [VERSION], synthetic India-calibrated cohort, params [HASH]
- 2 S01 - WHO, Guidelines on second- and third-line medicines and type of insulin for the control of blood glucose levels in non-pregnant adults with diabetes mellitus, 2018, § [SECTION]
- 3 [SOURCE] for rules R-[ID], [VERSION], § [SECTION]

i The clinician decides.

Ask about these three options for this patient



Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

SGLT2 inhibitor

added to metformin

👍 Safe to consider

†† More HbA1c lowering than DPP-4i

HBA1C CHANGE AT 6 MONTHS

-1.05 % 95% CI -1.22 to -0.88



VS DPP-4I

-0.21 % 95% CI -0.36 to -0.06

- ✓ Interval excludes 0

HYPOGLYCAEMIA RISK

●○○ Low

WEIGHT CHANGE AT 6 MONTHS

-2.1 kg 95% CI -2.6 to -1.6

COST

₹[MIN]–[MAX] per month

EVIDENCE LEVEL

Moderate

DPP-4 inhibitor

added to metformin

☑ **Safe to consider**

HBA1C CHANGE AT 6 MONTHS

-0.84 % 95% CI -0.99 to -0.69



VS DPP-4I

Comparator DPP-4 inhibitor is the reference

HYPOGLYCAEMIA RISK

●○○ Low

WEIGHT CHANGE AT 6 MONTHS

-0.2 kg 95% CI -0.6 to +0.2

COST

Phone 390 • part 3 of 6

₹[MIN]–[MAX] per month

EVIDENCE LEVEL

Moderate

Sulfonylurea

added to metformin

⊗

Do not use

HBA1C CHANGE AT 6 MONTHS

Not estimated

Removed by rule R-[ID] before estimation

VS DPP-4I

—

HYPOGLYCAEMIA RISK

—

WEIGHT CHANGE AT 6 MONTHS

—

COST

₹[MIN]–[MAX] per month

EVIDENCE LEVEL

Not assessed

Evidence level — Moderate: a narrow interval and enough similar patients. Low: a wide interval or few similar patients. Insufficient: no estimate is shown. It is never “High”, because every estimate comes from a synthetic cohort.

Trade-offs

• **Weight:** SGLT2 inhibitor –2.1 kg (95% CI –2.6 to –1.6); DPP-4 inhibitor –0.2 kg (95% CI –0.6 to +0.2). 1

• **Hypoglycaemia:** both remaining options are in the low-risk category. 2

Options are in the top-left category.

- **Cost:** compare each option against the patient's budget of ₹[AMOUNT] per month.

What drives this estimate

Difference between SGLT2 inhibitor and DPP-4 inhibitor for this patient. Top 3 drivers, in HbA1c points.

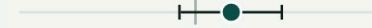
Higher starting HbA1c (this patient 8.4%): 0.04 more lowering (95% CI 0.01 to 0.07)



eGFR 62 mL/min/1.73m²: 0.03 less lowering (95% CI 0.00 to 0.06)



BMI 31.2 kg/m²: 0.02 more lowering (95% CI -0.01 to 0.05)



← less lowering 0 more lowering →

These show how the estimate changes across patients like this one. They are not causes.

Sources cited

- 1 DiaCausal causal engine [VERSION], synthetic India-calibrated cohort, params [HASH]
- 2 S01 · WHO, Guidelines on second- and

third-line medicines and type of insulin for the control of blood glucose levels in non-pregnant adults with diabetes mellitus, 2018, § [SECTION]

3 [SOURCE] for rules R-[ID], [VERSION], § [SECTION]



The clinician decides.

Ask about these three options for this patient



Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

Patient Details

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18-evidence-drawer

- Evidence drawer

fixed in P05

Adds page and licence; S01 WHO 2018 metadata; opens as a side sheet over 17 (bottom sheet on a phone).

Desktop 1280 · part 1 of 3

DiaCausal

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

Patient DetailsInvestigateAnalysisGuideAbout

Patient details

Example data

New patient

Clears these details and the conversation.

Age

18 or older.

58years

Diabetes duration

6years

HbA1c

8.4%

eGFR

62mL/min/1.73m²

BMI

31.2kg/m²

Category: Obese (≥25)

Adult, 58 y · T2D 6 y · on metformin · HbA1c 8.4% · eGFR 62 mL/min/1.73 m²

Example data

YOU ASKED

Compare the three options for this patient.

DIACAUSAL ANSWERED

Three options compared for this patient

Safety rules ran first. One of the three options was rejected.

Safety checks

SGLT2 inhibitor

Safe to consider

No exclusion rule matched.
eGFR 62 is above this rule's kidney threshold.

Rule R-[ID] · checked on eGFR,
type 1 diabetes, past ketoacidosis

Source [SOURCE], [VERSION], § [SECTION]

DPP-4 inhibitor

Safe to consider

No exclusion rule matched this record.

Rule R-[ID] · checked on pancreatitis, heart failure

Source [SOURCE], [VERSION], § [SECTION]

EVIDENCE 2

WHO guideline on second- and third-line medicines

Source

S01 · WHO, Guidelines on second- and third-line medicines and type of insulin for the control of blood glucose levels in non-pregnant adults with diabetes mellitus

Version

2018 · ISBN 978-92-4-155028-4

Section

§ [SECTION NUMBER AND HEADING]

Page

[PAGE]

Licence

[LICENCE, AS RECORDED IN THE LICENCE TABLE]

Retrieved

[DATE] · index [INDEX VERSION]

Retrieved passage

[PASSAGE TEXT EXACTLY AS STORED IN THE INDEX — QUOTED, NOT PARAPHRASED.]

Shown exactly as stored. If it does not support the claim it is attached to, report it.

Q Open in Investigate

Report a mismatch

Phone 390 • part 1 of 6

DiaCausal

Sign out

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

PATIENT DETAILS

Adult, 58 y · T2D 6 y · on metformin · HbA1c 8.4% · eGFR 62 · BMI 31.2 · past hypoglycaemia: severe

Example data

YOU ASKED

Compare the three options for this patient.

DIACAUSAL ANSWERED

Example values

Three options compared for this patient

Safety rules ran first. One of the three options was removed before any estimate was made.

Safety checks

SGLT2 inhibitor

Safe to consider

No exclusion rule matched. eGFR 62 is above this rule's kidney threshold.

Rule R-[ID] · checked on eGFR, type 1 diabetes, past ketoacidosis

Desktop 1280 · part 2 of 3

Asian-Indian cut-offs: normal <23 · overweight 23–24.9 · obese ≥25

Established heart disease

✓ Yes

No

Chronic kidney disease

Yes

✓ No

Heart failure

Yes

✓ No

Past hypoglycaemia

None

Mild

✓ Severe

Budget for this drug

What the patient can spend each month.

₹ [AMOUNT]

₹ / month

HbA1c at 6 months

↑↑ SGLT2 inhibitor shows more HbA1c lowering than DPP-4i difference −0.21 % (95% CI −0.36 to −0.06). The interval excludes zero. This is a difference on HbA1c only. 1

OPTION	HBA1C CHANGE, 6 MONTHS	DIFFERENCE VS DPP-4I	HYP GLY
SGLT2 inhibitor added to metformin	−1.05 % 95% CI −1.22 to −0.88	−0.21 % 95% CI −0.36 to −0.06	●○○○
Safe to consider		Interval excludes 0	
↑↑ More HbA1c lowering than DPP-4i			
DPP-4 inhibitor added to metformin	−0.84 % 95% CI −0.99 to −0.69	Comparator	●○○○
Safe to consider		DPP-4 inhibitor is the reference	
Sulfonylurea added to metformin	Not estimated Removed by rule R-[ID] before estimation	—	—
Do not use			

Evidence level — Moderate: a narrow interval and enough similar patients. Insufficient: no estimate is shown. It is never “High”, because every evidence option has a narrow interval.

Trade-offs

- Weight: SGLT2 inhibitor −2.1 kg (95% CI −2.6 to −1.6); DPP-4i −0.5 kg (95% CI −1.0 to 0.0). 1
- Hypoglycaemia: both remaining options are in the low-risk group.
- Cost: compare each option against the patient's budget.

Phone 390 · part 2 of 6

Source [SOURCE], [VERSION], § [SECTION] 3

DPP-4 inhibitor

✓ Safe to consider

No exclusion rule matched for this record.

Rule R-[ID] · checked on past pancreatitis, heart failure

Source [SOURCE], [VERSION], § [SECTION] 3

Sulfonylurea

✗ Do not use

Removed before estimation. It is not compared with the others.

Rule R-[ID] · triggered by past hypoglycaemia: severe

Source [SOURCE], [VERSION], § [SECTION] 3

HbA1c at 6 months

↑↑ SGLT2 inhibitor shows more HbA1c lowering than DPP-4 inhibitor for this patient: difference −0.21 % (95% CI −0.36 to −0.06). The interval excludes zero. This is a difference on HbA1c only. 1

18-evidence-drawer

- Evidence drawer

fixed in P05

Desktop 1280 · part 3 of 3

The image displays a research prototype for clinician evaluation, featuring a patient summary, a decision tree, and a list of sources cited.

Patient Summary:

- Age:** 60 years
- Sex:** Male
- Weight:** 70 kg
- Height:** 1.73 m
- Diagnosis:** Type 2 Diabetes Mellitus
- Current Medication:** Metformin 1000 mg daily
- Current HbA1c:** 8.4%
- Current eGFR:** 62 mL/min/1.73m²
- Current BMI:** 31.2 kg/m²

What drives this estimate

Difference between SGLT2 inhibitor and DPP-4 inhibitor for this patient

Higher starting HbA1c (this patient 8.4%): 0.04 more lowering (95% CI 0.01 to 0.07)

eGFR 62 mL/min/1.73m²: 0.03 less lowering (95% CI 0.00 to 0.06)

BMI 31.2 kg/m²: 0.02 more lowering (95% CI -0.01 to 0.05)

These show how the estimate changes across patients like this one

Sources cited

- 1 DiaCausal causal engine [VERSION], synthetic India-calibrate
- 2 S01 - WHO, Guidelines on second- and third-line medicines and in non-pregnant adults with diabetes mellitus, 2018, § [SECTION]
- 3 [SOURCE] for rules R-[ID], [VERSION], § [SECTION]

The clinician decides.

Ask about these three options for this patient

Research prototype for clinician evaluation; not a marketed medical device

Phone 390 • part 3 of 6

SGLT2 inhibitor added to metformin ✓ Safe to consider ↑‡ More HbA1c lowering than DPP-4i HbA1c CHANGE AT 6 MONTHS -1.05 % 95% CI -1.22 to -0.88 VS DPP-4I -0.21 % 95% CI -0.36 to -0.06 ✓ Interval excludes 0 HYPOGLYCAEMIA RISK ●○○ Low WEIGHT CHANGE AT 6 MONTHS -2.1 kg 95% CI -2.6 to -1.6 COST ₹[MIN]-[MAX] per month EVIDENCE LEVEL ■ Moderate	DPP-4 inhibitor added to metformin ✓ Safe to consider HbA1c CHANGE AT 6 MONTHS -0.84 % 95% CI -0.99 to -0.69 VS DPP-4I Comparator DPP-4 inhibitor is the reference HYPOGLYCAEMIA RISK ●○○ Low WEIGHT CHANGE AT 6 MONTHS -0.2 kg 95% CI -0.6 to +0.2 COST
--	--

₹[MIN]–[MAX] per month

EVIDENCE LEVEL

Moderate

Sulfonylurea

added to metformin

⊗

Do not use

HBA1C CHANGE AT 6 MONTHS

Not estimated Removed by rule R-[ID] before estimation

VS DPP-4I

—

HYPOGLYCAEMIA RISK

—

WEIGHT CHANGE AT 6 MONTHS

—

COST

₹[MIN]–[MAX] per month

EVIDENCE LEVEL

Not assessed

Evidence level — Moderate: a narrow interval and enough similar patients. Low: a wide interval or few similar patients. Insufficient: no estimate is shown. It is never “High”, because every estimate comes from a synthetic cohort.

Trade-offs

- Weight:** SGLT2 inhibitor −2.1 kg (95% CI −2.6 to −1.6); DPP-4 inhibitor −0.2 kg (95% CI −0.6 to +0.2). ①
- Hypoglycaemia:** both remaining options are in the low-risk category. ②

Options are in the top-right category.

- **Cost:** compare each option against the patient's budget of ₹[AMOUNT] per month.

What drives this estimate

Difference between SGLT2 inhibitor and DPP-4 inhibitor for this patient. Top 3 drivers, in HbA1c points.

Higher starting HbA1c (this patient 8.4%): 0.04 more lowering (95% CI 0.01 to 0.07)



EVIDENCE 2

×

WHO guideline on second- and third-line medicines

Source	S01 · WHO, Guidelines on second- and third-line medicines and type of insulin for the control of blood glucose levels in non-pregnant adults with diabetes mellitus
Version	2018 · ISBN 978-92-4-155028-4
Section	§ [SECTION NUMBER AND HEADING]
Page	[PAGE]
Licence	[LICENCE, AS RECORDED IN THE LICENCE TABLE]

Retrieved [DATE] · index [INDEX VERSION]

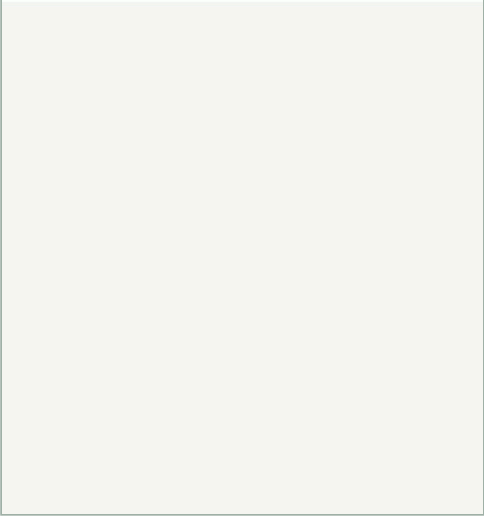
Retrieved passage

[PASSAGE TEXT EXACTLY AS STORED
IN THE INDEX — QUOTED, NOT
PARAPHRASED.]

Shown exactly as stored. If it does not support the
claim it is attached to, report it.

 **Open in Investigate**

Report a mismatch



19-insufficient-evidence · Insufficient evidence fixed in P05

Calmer wording; reasons tied to the 95% interval; points to Investigate; ends “The clinician decides.”

Desktop 1280 · part 1 of 2

DiaCausal

[USER NAME] Sign out

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

Patient DetailsInvestigateAnalysisGuideAbout

Patient detailsExample data

New patient
Clears these details and the conversation.
Age
18 or older.
58years
Diabetes duration
6years
HbA1c
8.4%
eGFR
62mL/min/1.73m²
BMI
31.2kg/m²
Category: Obese (≥25)

Adult, 58 y · T2D 6 y · on metformin · HbA1c 8.4% · eGFR 62 · BMI 31.2Example data
YOU ASKED
Compare the three options for this patient.
DIACAUSAL ANSWEREDExample
Insufficient evidence — no comparison shown

This patient is outside the range the model has seen, so DiaCausal will not show estimates for this record.

Why

[FIELD] is outside the range covered by the synthetic cohort ([RANGE]).

Too few similar patients received each option to estimate a usable 95% interval.

Only the reasons that applied are listed. 1

What you can do instead

Check the patient details for a typing error, then compare again.

Look up the question in Investigate, which shows cited passages without estimates.

Phone 390 • part 1 of 2

DiaCausal

Sign out

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

PATIENT DETAILS

Adult, 58 y · T2D 6 y · on metformin · HbA1c 8.4% · eGFR 62 · BMI 31.2

Example data

YOU ASKED

Compare the three options for this patient.

DIACAUSAL ANSWERED

Example

Insufficient evidence — no comparison shown

This patient is outside the range the model has seen, so DiaCausal will not show estimates for this record.

Why

[FIELD] is outside the range covered by the synthetic cohort ([RANGE]).

Too few similar patients received each option to estimate a usable 95% interval.

Only the reasons that applied are listed. 1

Desktop 1280 · part 2 of 2

Asian-Indian cut-offs: normal <23 · overweight 23–24.9 · obese ≥25

Established heart disease

✓ Yes

No

Chronic kidney disease

Yes

✓ No

Heart failure

Yes

✓ No

Past hypoglycaemia

✓ None

Mild

Severe

Budget for this drug

What the patient can spend each month.

₹ [AMOUNT]

₹ / month

📖 Use your usual guideline, as you would without this tool.

📘 The clinician decides.

Ask about these three options for this patient

↑

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

Phone 390 · part 2 of 2

What you can do instead

🔄 Check the patient details for a typing error, then compare again.

🔍 Look up the question in Investigate, which shows cited passages without estimates.

📖 Use your usual guideline, as you would without this tool.

📘 The clinician decides.

Ask about these three options for this patient

↑

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

📖 Patient Details

🔍 Investigate

📊 Analysis

📌 Guide

👤 About

20-loading-stages · Working through it fixed in P05

Six stages reordered to the pipeline: input checks, safety rules, causal estimate, retrieval, writing, output checks.

Desktop 1280 · part 1 of 2

DiaCausal

[USER NAME]

Sign out

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

Patient Details

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About

Patient details

Example data

New patient

Clears these details and the conversation.

Age

18 or older.

58

years

Diabetes duration

6

years

HbA1c

8.4

%

eGFR

62

mL/min/1.73m²

BMI

31.2

kg/m²

Category: Obese (≥25)

Adult, 58 y · T2D 6 y · on metformin · HbA1c 8.4% · eGFR 62 · BMI 31.2

Example data

YOU ASKED

Compare the three options for this patient.

Working through it

Six steps run in order. Safety rules finish before anything is estimated.

Checking the question and patient details

Done

Applying safety rules

Done

Estimating each option's HbA1c change with a 95% interval

Running

Retrieving cited passages

Waiting

Writing the answer (local model, or template if it is unavailable)

Waiting

Checking the answer before it is shown

Waiting

Step 3 of 6.

Phone 390 · part 1 of 2

DiaCausal

Sign out

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

PATIENT DETAILS

▼

Adult, 58 y · T2D 6 y · on metformin · HbA1c 8.4% · eGFR 62 · BMI 31.2

Example data

YOU ASKED

Compare the three options for this patient.

Working through it

Six steps run in order. Safety rules finish before anything is estimated.

Checking the question and patient details

Done

Applying safety rules

Done

Estimating each option's HbA1c change with a 95% interval

Running

Retrieving cited passages

Waiting

Desktop 1280 · part 2 of 2

Asian-Indian cut-offs: normal <23 · overweight 23-24.9 · obese ≥25

Established heart disease

✓ Yes

No

Chronic kidney disease

Yes

✓ No

Heart failure

Yes

✓ No

Past hypoglycaemia

✓ None

Mild

Severe

Budget for this drug

What the patient can spend each month.

₹ [AMOUNT]

₹ / month

Ask about these three options for this patient

↑

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

Phone 390 · part 2 of 2

Writing the answer (local model, or template if it is unavailable)

Waiting

Checking the answer before it is shown

Waiting

Step 3 of 6.

Ask about these three options for this patient

↑

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

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21-answer-no-clear-difference

- No clear difference

new

Both intervals vs DPP-4i include 0, so no leader; trade-offs come first; sulfonylurea is “Check first”.

Desktop 1280 · part 1 of 3

[USER NAME] [Sign out](#)

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

[Patient Details](#)
[Investigate](#)
[Analysis](#)
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Patient details

[Example data](#)

New patient

Clears these details and the conversation.

Age

18 or older.

58 years

Diabetes duration

6 years

HbA1c

8.4 %

eGFR

62 mL/min/1.73m²

BMI

31.2 kg/m²

Category: Obese (≥ 25)

Adult, 58 y · T2D 6 y · on metformin · HbA1c 8.4% · eGFR 62 · BMI 31.2 · past hypoglycaemia: mild

[Example data](#)

YOU ASKED

Compare the three options for this patient.

DIACAUSAL ANSWERED [Example values](#)

Three options compared for this patient

✓ Safety rules ran first. No option was removed; one needs checking.

Safety checks

SGLT2 inhibitor	DPP-4 inhibitor	Sulfonylurea
✓ Safe to consider No exclusion rule matched. eGFR 62 is above this rule's kidney threshold.	✓ Safe to consider No exclusion rule matched for this record.	⚠ Check first Can be compared, but confirm the hypoglycaemia history before relying on it.
Rule R-[ID] · checked on eGFR, type 1 diabetes, past ketoacidosis Source [SOURCE], [VERSION], § [SECTION] ③	Rule R-[ID] · checked on past pancreatitis, heart failure Source [SOURCE], [VERSION], § [SECTION] ③	Rule R-[ID] · triggered by past hypoglycaemia: mild Source [SOURCE], [VERSION], § [SECTION] ③

Phone 390 • part 1 of 6

DiaCausal

[Sign out](#)

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

PATIENT DETAILS

Adult, 58 y · T2D 6 y · on metformin · HbA1c 8.4% · eGFR 62 · BMI 31.2 · past hypoglycaemia: mild

① Example data

YOU ASKED

Compare the three options for this patient.

DIACAUSAL ANSWERED

① Example values

Three options compared for this patient

Safety rules ran first. No option was removed; one needs checking.

Safety checks

SGLT2 inhibitor

Safe to consider

No exclusion rule matched. eGFR 62 is above this rule's kidney threshold.

Rule R-[ID] · checked on eGFR, type 1 diabetes, past ketoacidosis

Asian-Indian cut-offs: normal <23 · overweight 23–24.9 · obese ≥25

Established heart disease

✓ Yes

No

Chronic kidney disease

Yes

✓ No

Heart failure

Yes

✓ No

Past hypoglycaemia

None

✓ Mild

Severe

Budget for this drug

What the patient can spend each month.

₹ [AMOUNT]

₹ / month

🚫 No clear difference in HbA1c for this patient.

Both differences against DPP-4 inhibitor have 95% intervals that include zero.

1

Trade-offs

• **Hypoglycaemia:** sulfonylurea is in the high-risk category, and this patient has had mild hypoglycaemia before.

2

• **Weight:** SGLT2 inhibitor −1.9 kg (95% CI −2.4 to −1.4); DPP-4 inhibitor −0.1 kg (95% CI −0.5 to +0.3); sulfonylurea +1.6 kg (95% CI +0.9 to +2.3).

1

• **Cost:** compare each option against the patient's budget of ₹[AMOUNT] per month.

HbA1c at 6 months

OPTION	HBA1C CHANGE, 6 MONTHS dot = estimate · bar = 95% CI · axis −1.6 to 0 %	DIFFERENCE VS DPP-4i	HYPO-GLYCAEMIA	WEIGHT, 6 MONTHS	COST	EVIDENCE LEVEL
SGLT2 inhibitor added to metformin 🟢 Safe to consider	−0.96 % 95% CI −1.14 to −0.78 ├─●─┤	−0.08 % 95% CI −0.27 to +0.11 ○ Interval includes 0	●○○ Low	−1.9 kg 95% CI −2.4 to −1.4	₹[MIN]–[MAX] per month	■ Moderate
DPP-4 inhibitor added to metformin 🟢 Safe to consider	−0.88 % 95% CI −1.03 to −0.73 ├─●─┤	Comparator DPP-4 inhibitor is the reference	●○○ Low	−0.1 kg 95% CI −0.5 to +0.3	₹[MIN]–[MAX] per month	■ Moderate
Sulfonylurea added to metformin ⚠ Check first	−1.02 % 95% CI −1.30 to −0.74 ├─●─┤	−0.14 % 95% CI −0.42 to +0.14 ○ Interval includes 0	●●● High	+1.6 kg 95% CI +0.9 to +2.3	₹[MIN]–[MAX] per month	■ Low

Evidence level

— Moderate: a narrow interval and enough similar patients. Low: a wide interval or few similar patients. Insufficient: no estimate is shown. It is never “High”, because every estimate comes from a synthetic cohort.

Phone 390 · part 2 of 6

Source [SOURCE], [VERSION], § [SECTION]

3

DPP-4 inhibitor

🟢 Safe to consider

No exclusion rule matched for this record.

Rule R-[ID] · checked on past pancreatitis, heart failure

Source [SOURCE], [VERSION], § [SECTION]

3

Sulfonylurea

⚠ Check first

Can be compared, but confirm the hypoglycaemia history before relying on it.

Rule R-[ID] · triggered by past hypoglycaemia: mild

Source [SOURCE], [VERSION], § [SECTION]

3

🚫 No clear difference in HbA1c for this patient.

Both differences against DPP-4 inhibitor have 95% intervals that include zero.

1

Trade-offs

• **Hypoglycaemia:** sulfonylurea is in the high-risk category, and this patient has

21-answer-no-clear-difference

- No clear difference

new

Desktop 1280 · part 3 of 3

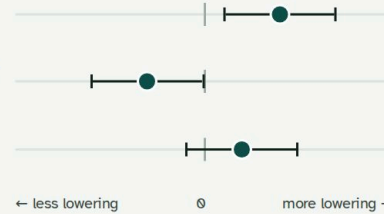
What drives this estimate

Difference between SGLT2 inhibitor and DPP-4 inhibitor for this patient. Top 3 drivers, in HbA1c points.

Higher starting HbA1c (this patient 8.4%): 0.04 more lowering (95% CI 0.01 to 0.07)

eGFR 62 mL/min/1.73m²: 0.03 less lowering (95% CI 0.00 to 0.06)

BMI 31.2 kg/m²: 0.02 more lowering (95% CI -0.01 to 0.05)



These show how the estimate changes across patients like this one. They are not causes.

Sources cited

- ① DiaCausal causal engine [VERSION], synthetic India-calibrated cohort, params [HASH]
- ② S01 - WHO, Guidelines on second- and third-line medicines and type of insulin for the control of blood glucose levels in non-pregnant adults with diabetes mellitus, 2018, § [SECTION]
- ③ [SOURCE] for rules R-[ID], [VERSION], § [SECTION]

① **The clinician decides.**

Ask about these three options for this patient



Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

Phone 390 • part 3 of 6

had mild hypoglycaemia before. (2)

- **Weight:** SGLT2 inhibitor -1.9 kg (95% CI -2.4 to -1.4); DPP-4 inhibitor -0.1 kg (95% CI -0.5 to +0.3); sulfonylurea +1.6 kg (95% CI +0.9 to +2.3). (1)
- **Cost:** compare each option against the patient's budget of ₹[AMOUNT] per month.

HbA1c at 6 months

SGLT2 inhibitor

added to metformin

✔ Safe to consider

HBA1C CHANGE AT 6 MONTHS

-0.96 % 95% CI -1.14 to -0.78



VS DPP-4I

-0.08 % 95% CI -0.27 to +0.11

☐ Interval includes 0

HYPOGLYCAEMIA RISK

●○○ Low

WEIGHT CHANGE AT 6 MONTHS

-1.9 kg 95% CI -2.4 to -1.4

COST

₹[MIN]–[MAX] per month

EVIDENCE LEVEL

Moderate

DPP-4 inhibitor

added to metformin

✔ Safe to consider

HbA1c change at 6 months
-0.88 % 95% CI -1.03 to -0.73



VS DPP-4I
Comparator DPP-4 inhibitor is the reference

Hypoglycaemia risk
●○○ Low

Weight change at 6 months
-0.1 kg 95% CI -0.5 to +0.3

Cost
₹[MIN]-[MAX] per month

Evidence level
■|| Moderate

Sulfonylurea
added to metformin

⚠ Check first

HbA1c change at 6 months
-1.02 % 95% CI -1.30 to -0.74



VS DPP-4I
-0.14 % 95% CI -0.42 to +0.14

○ Interval includes 0

Hypoglycaemia risk
●●● High

Weight change at 6 months
+1.6 kg 95% CI +0.9 to +2.3

Cost
₹[MIN]-[MAX] per month

Evidence level
■|| Low

Evidence level — Moderate: a narrow interval and

enough similar patients. Low: a wide interval or few similar patients. Insufficient: no estimate is shown. It is never “High”, because every estimate comes from a synthetic cohort.

What drives this estimate

Difference between SGLT2 inhibitor and DPP-4 inhibitor for this patient. Top 3 drivers, in HbA1c points.

Higher starting HbA1c (this patient 8.4%): 0.04 more lowering (95% CI 0.01 to 0.07)



eGFR 62 mL/min/1.73m²: 0.03 less lowering (95% CI 0.00 to 0.06)



BMI 31.2 kg/m²: 0.02 more lowering (95% CI -0.01 to 0.05)



← less lowering 0 more lowering →

These show how the estimate changes across patients like this one. They are not causes.

Sources cited

① DiaCausal causal engine [VERSION], synthetic India-calibrated cohort, params [HASH]

② S01 · WHO, Guidelines on second- and

Phone 390 · part 6 of 6

third-line medicines and type of insulin for the control of blood glucose levels in non-pregnant adults with diabetes mellitus, 2018, § [SECTION]

3 [SOURCE] for rules R-[ID], [VERSION], § [SECTION]



The clinician decides.

Ask about these three options for this patient



Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.



Patient Details



Investigate



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About

Search box, example searches, sources list, cited passages with source, version, section, page and licence.

Desktop 1280 · part 1 of 3

⌵ DiaCausal

[USER NAME] [Sign out](#)

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

Patient Details

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Investigate

Search the licence-cleared sources directly. Every passage is shown as stored, with its source, version, section, page and licence. Nothing here is generated.

Search the evidence

SGLT2 inhibitor kidney function

Q Search

Metformin and kidney function

SGLT2 inhibitors and ketoacidosis

DPP-4 inhibitors and heart failure

Sulfonylureas and hypoglycaemia

Sources in the search

S01 · WHO, 2018

Guidelines on second- and third-line medicines and type of insulin...

✓ In the search

S0[N] · US FDA

Drug Safety Communications, [DATES]

3 passages found · ranked by keyword and TF-IDF search, fused ·

Example layout

1 Guidelines on second- and third-line medicines...

Source

S01 · WHO

Version

2018 · ISBN 978-92-4-155028-4

Section

§ [SECTION]

Page

[PAGE]

Licence

[LICENCE]

Phone 390 · part 1 of 4

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Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

Investigate

Search the licence-cleared sources directly. Every passage is shown as stored, with its source, version, section, page and licence. Nothing here is generated.

Search the evidence

SGLT2 inhibitor kidney function

Q Search

Metformin and kidney function

SGLT2 inhibitors and ketoacidosis

DPP-4 inhibitors and heart failure

Sulfonylureas and hypoglycaemia

Sources in the search

3 passages found · ranked by keyword and TF-

✔ In the search

S0[N] · [SOURCE]
[TITLE]

🕒 Cleared, not yet added

Only licence-cleared sources are searched. [N] passages in the index.

[PASSAGE TEXT EXACTLY AS STORED IN THE INDEX – QUOTED, NOT PARAPHRASED.]

Copy citation

[Open source](#)

② **Drug Safety Communication: [TITLE]**

Source S0[N] · US FDA

Version [DATE]

Section § [SECTION]

Page [PAGE]

Licence [LICENCE]

[PASSAGE TEXT EXACTLY AS STORED IN THE INDEX — QUOTED, NOT PARAPHRASED.]

Copy citation

[Open source](#)

3 Guidelines on second- and third-line medicines...

Source S01 . WHO

Version 2018 · ISBN 978-92-4-155028-4

Section § [SECTION]

Page [PAGE]

Licence [LICENCE]

IDF search, fused .

Example layout

1

Guidelines on second- and third-line medicines...

Source S01 - WHO

Version 2018 · ISBN 978-92-4-155028-4

Section § [SECTION]

Page [PAGE]

Licence [LICENCE]

[PASSAGE TEXT EXACTLY AS
STORED IN THE INDEX — QUOTED,
NOT PARAPHRASED.]

 **Copy citation**

 **Open source**

2

Drug Safety Communication:
[TITLE]

Source	S0[N] · US FDA
--------	----------------

Version [DATE]

Section § [SECTION]

Page [PAGE]

[PASSAGE TEXT EXACTLY AS STORED IN THE INDEX — QUOTED, NOT PARAPHRASED.]

Copy citation

Open source

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

Licence [LICENCE]

[PASSAGE TEXT EXACTLY AS STORED IN THE INDEX — QUOTED, NOT PARAPHRASED.]

Copy citation

Open source

3

Guidelines on second- and third-line medicines...

Source S01 · WHO

Version 2018 · ISBN 978-92-4-155028-4

Section § [SECTION]

Page [PAGE]

Licence [LICENCE]

[PASSAGE TEXT EXACTLY AS STORED IN THE INDEX — QUOTED, NOT PARAPHRASED.]

Copy citation

Open source

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

medical device, not for unsupervised clinical use.



Patient
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About

Insufficient-evidence state. No generated answer.

Desktop 1280 · part 1 of 2

DiaCausal

[USER NAME] [Sign out](#)

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

Patient Details

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Investigate

Search the licence-cleared sources directly. Every passage is shown as stored, with its source, version, section, page and licence. Nothing here is generated.

Search the evidence

[QUERY THE SOURCES DO NOT COVER]

Q Search

Metformin and kidney function

SGLT2 inhibitors and ketoacidosis

DPP-4 inhibitors and heart failure

Sulfonylureas and hypoglycaemia

Sources in the search

S01 · WHO, 2018

Guidelines on second- and third-line medicines and type of insulin...

✔ In the search

S0[N] · US FDA

Drug Safety Communications, [DATES]

⚠ Insufficient evidence. No passage in the licence-cleared sources answers this search.

Why this can happen

- The topic is not covered by the sources currently in the search.
- The search words differ from the wording the source uses.

What you can do

Try different words or one of the example searches above

Phone 390 · part 1 of 2

DiaCausal

[Sign out](#)

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

Investigate

Search the licence-cleared sources directly. Every passage is shown as stored, with its source, version, section, page and licence. Nothing here is generated.

Search the evidence

[QUERY THE SOURCES DO NOT

Q Search

Metformin and kidney function

SGLT2 inhibitors and ketoacidosis

DPP-4 inhibitors and heart failure

Sulfonylureas and hypoglycaemia

Sources in the search

23-investigate-insufficient

- Investigate — no passage

new

Desktop 1280 · part 2 of 2

✔ In the search

S0[N] · [SOURCE]

[TITLE]

🕒 Cleared, not yet added

Only licence-cleared sources are searched. [N] passages in the index.

- Try different words, or one of the example searches above.
- Check the list of sources to see what is covered.
- Use your usual guideline. DiaCausal does not write an answer when it has no source.

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use

Phone 390 • part 2 of 2



Insufficient evidence. No passage in the licence-cleared sources answers this search.

Why this can happen

- The topic is not covered by the sources currently in the search.
- The search words differ from the wording the source uses.

What you can do

- Try different words, or one of the example searches above.
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- Use your usual guideline. DiaCausal does not write an answer when it has no source.

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

Investigat


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Versions A–D table, SHAP A vs C, LIME stability. Labelled synthetic benchmark, example values.

Desktop 1280 · part 1 of 3

⌵ DiaCausal

[USER NAME] [Sign out](#)

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

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Synthetic benchmark · example values

How four versions of DiaCausal compare on a synthetic India-calibrated benchmark where the true effects are known. This tests whether the methods recover a planted truth. It says nothing about real-world drug effects.

Versions A–D compared

VERSION	USES	POLICY REGRET (HbA1c POINTS, LOWER IS BETTER)	95% INTERVAL COVERAGE	EXPLANATION STABILITY	CLAIMS WITH A CITATION	ANSWERS WITHHELD
A XAI only	SHAP + LIME on a model that predicts HbA1c	0.09 95% CI 0.07–0.11	No interval	0.62 95% CI 0.55–0.69	None	Never
B Causal only	Safety rules + causal engine	0.011 95% CI 0.008–0.014	94% 95% CI 91–96	No explanation	None	2.5% 95% CI 2.1–2.9
C Causal + XAI	B + SHAP on the causal estimate	0.011 95% CI 0.008–0.014	94% 95% CI 91–96	0.88 95% CI 0.84–0.92	None	2.5% 95% CI 2.1–2.9
D Causal + XAI + RAG	C + cited passages	0.011 95% CI 0.008–0.014	94% 95% CI 91–96	0.88 95% CI 0.84–0.92	100% 95% CI 96–100	2.5% 95% CI 2.1–2.9

Policy regret: how much HbA1c lowering is lost by following each version's ordering instead of the true best option. Explanation stability: agreement of the top-3 features across 30 reruns (0–1). A has no interval, so its coverage cannot be measured. No version is marked “best”: the table is for comparison, not ranking.

Phone 390 · part 1 of 6

⌵ DiaCausal

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Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

Analysis

Synthetic benchmark · example values

How four versions of DiaCausal compare on a synthetic India-calibrated benchmark where the true effects are known. This tests whether the methods recover a planted truth. It says nothing about real-world drug effects.

Versions A–D compared

A XAI only

USES
SHAP + LIME on a model that predicts HbA1c

POLICY REGRET (HbA1c POINTS, LOWER IS BETTER)
0.09 95% CI 0.07–0.11

95% INTERVAL COVERAGE
No interval

EXPLANATION STABILITY
0.62 95% CI 0.55–0.69

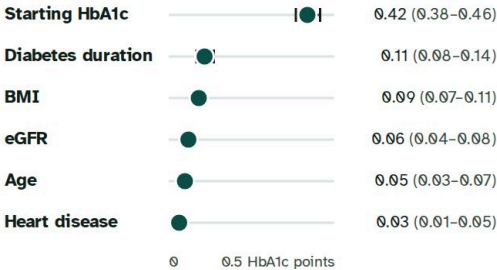
CLAIMS WITH A CITATION
None

ANSWERS WITHHELD
Never

B Causal only

A · SHAP on the HbA1c prediction model

Explains what the prediction model uses to predict HbA1c. It does not explain treatment effects.

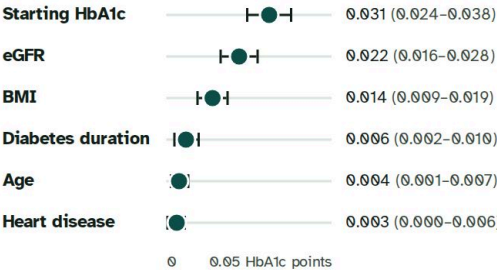


Scales differ between A and C: A explains a predicted HbA1c level; C explains a difference between two options. Compare the order, not the bar lengths.

Show values as a table

C · SHAP on the causal estimate

SGLT2 inhibitor vs DPP-4 inhibitor. The estimate is larger or smaller for patients with these features — these are not causes.



Mean absolute SHAP value, with a bootstrap 95% CI.

Show values as a table

LIME stability across reruns

Share of the top-3 features that stay the same across 30 reruns with different random seeds (0 = none, 1 = all). Dot = estimate, bar = 95% CI.



Show values as a table

USES

Safety rules + causal engine

POLICY REGRET (HBA1C POINTS, LOWER IS BETTER)

0.011 95% CI 0.008–0.014

95% INTERVAL COVERAGE

94% 95% CI 91–96

EXPLANATION STABILITY

No explanation

CLAIMS WITH A CITATION

None

ANSWERS WITHHELD

2.5% 95% CI 2.1–2.9

C Causal + XAI

USES

B + SHAP on the causal estimate

POLICY REGRET (HBA1C POINTS, LOWER IS BETTER)

0.011 95% CI 0.008–0.014

95% INTERVAL COVERAGE

94% 95% CI 91–96

EXPLANATION STABILITY

0.88 95% CI 0.84–0.92

CLAIMS WITH A CITATION

None

ANSWERS WITHHELD

2.5% 95% CI 2.1–2.9

D Causal + XAI + RAG

USES

C + cited passages

POLICY REGRET (HBA1C POINTS, LOWER IS BETTER)

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0.011 95% CI 0.008-0.014

95% INTERVAL COVERAGE

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CLAIMS WITH A CITATION

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A · SHAP on the HbA1c prediction model

Explains what the prediction model uses to predict HbA1c. It does not explain treatment effects.

Starting HbA1c



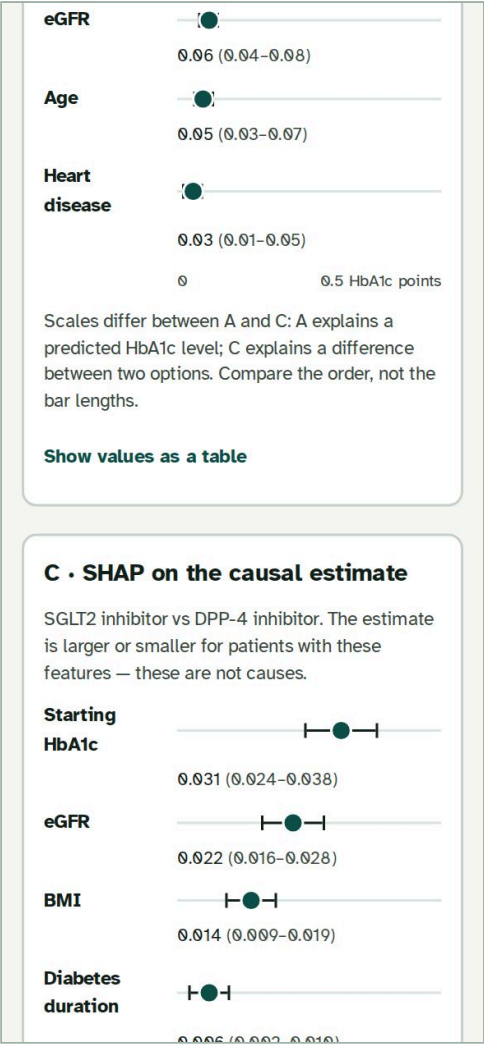
0.42 (0.38–0.46)

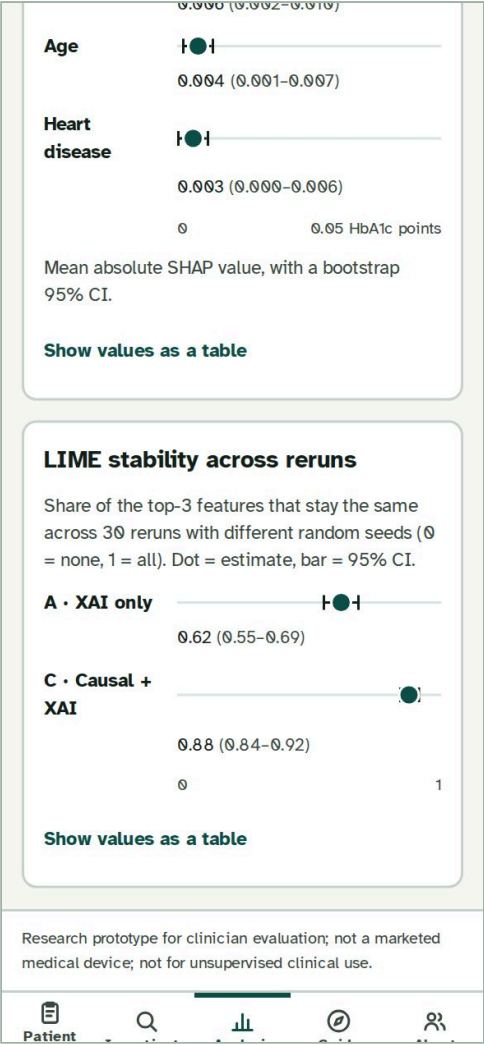
Diabetes duration



0.11 (0.08–0.14)

BMI 
0.09 (0.07-0.11)





Details	Investigate	Analysis	Guide	About

Intro, problem statement, solution, workflow, cautions, limitations, advantages.

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⌵ DiaCausal

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Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

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Guide

What DiaCausal is for, how it works, and where it stops.

Intro

Problem statement

Solution

Workflow

Cautions

Limitations

Advantages

Intro

DiaCausal is a decision-support research prototype for one decision: an adult with type 2 diabetes is already taking metformin, their HbA1c is still above target, and the doctor is choosing a second medicine. It compares three options — an SGLT2 inhibitor, a DPP-4 inhibitor and a sulfonylurea — for that one patient.

Problem statement

Average results from trials do not say how a particular patient is likely to respond, and looking at which patients did well on which drug in routine records mixes up the effect of the drug with the reasons it was chosen. A tool that simply predicts “who does well” can therefore point the wrong way. It also needs to show how sure it is and where its statements come from.

Solution

- **Safety rules first.** Cited rules remove options that are unsafe for this patient before anything is estimated.

Phone 390 · part 1 of 5

⌵ DiaCausal

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Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

Guide

What DiaCausal is for, how it works, and where it stops.

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Intro

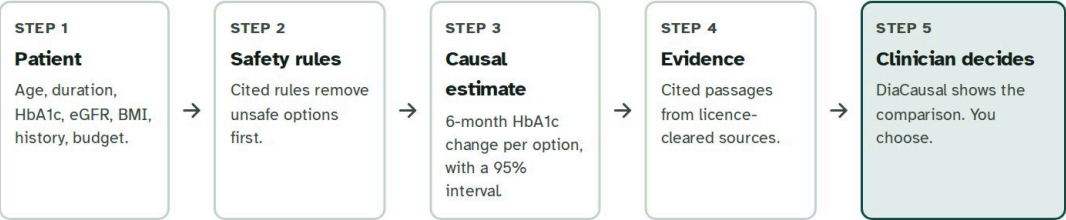
DiaCausal is a decision-support research prototype for one decision: an adult with type 2 diabetes is already taking metformin, their HbA1c is still above target, and the doctor is choosing a second medicine. It compares three options — an SGLT2 inhibitor, a DPP-4 inhibitor and a sulfonylurea — for that one patient.

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Average results from trials do not say how a particular patient is likely to respond,

- **Causal estimate.** A causal model estimates each remaining option's 6-month HbA1c change for this patient, with a 95% interval, correcting for why patients received each drug.
- **Evidence.** Passages from licence-cleared sources are retrieved and cited with source, version and section.
- **Abstain when unsure.** If the patient is unlike anyone the model has seen, it says “insufficient evidence” instead of guessing.

Workflow



Cautions

- Decision support only. DiaCausal never chooses for you.
- Check every value in the patient details before comparing. A typing error changes the estimate.
- Do not type names, Aadhaar, phone, PAN or email.
- In an emergency, follow your emergency protocol. DiaCausal will not answer.
- Doses are never shown. Use the product label and your formulary.

Limitations

- All estimates come from a synthetic India-calibrated cohort. They have not been validated on real patients.

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Workflow

STEP 1

- Only three options and one primary outcome (6-month HbA1c), with hypoglycaemia risk, weight change and monthly cost as secondary measures.
- The evidence search covers a small set of licence-cleared sources ([N] passages).
- Evidence level is never “High”, because the data are synthetic.
- Not covered: type 1 diabetes, pregnancy, under-18s, starting insulin.

Advantages

- Unsafe options are removed by cited, deterministic rules — not by a language model.
- Every estimate carries a 95% interval, and the tool withholds an answer rather than guess.
- Every clinical statement links to its source, version and section.
- Every request is logged and versioned, without patient values in the log.
- Works on a desktop monitor and on a phone.

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

Patient	
----------------	--

Age, duration, HbA1c, eGFR, BMI, history, budget.



STEP 2

Safety rules

Cited rules remove unsafe options first.



STEP 3

Causal estimate

6-month HbA1c change per option, with a 95% interval.



STEP 4

Evidence

Cited passages from licence-cleared sources.



STEP 5

Clinician decides

DiaCausal shows the comparison. You choose.

Cautions

- Decision support only. DiaCausal never chooses for you.
- Check every value in the patient details

before comparing. A typing error changes the estimate.

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Phone 390 · part 5 of 5

...deterministic rules — not by a language model.

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Four members with email placeholders, the guide, B.Tech Computer Engineering, FCRIT.

Desktop 1280 • part 1 of 2

DiaCausal

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About

Team

Group 28 • B.Tech Computer Engineering, FCRIT (Fr. C. Rodrigues Institute of Technology, Vashi) • 2026–27

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Advik Saxena

[EMAIL]

Guide

Mr. Rahul Jadhav

Project guide, Department of Computer Engineering, FCRIT

[EMAIL]

Engine [VERSION] • params [HASH] • rules [HASH] • synthetic cohort n=[N] seed=[SEED]

Phone 390 • part 1 of 2

DiaCausal

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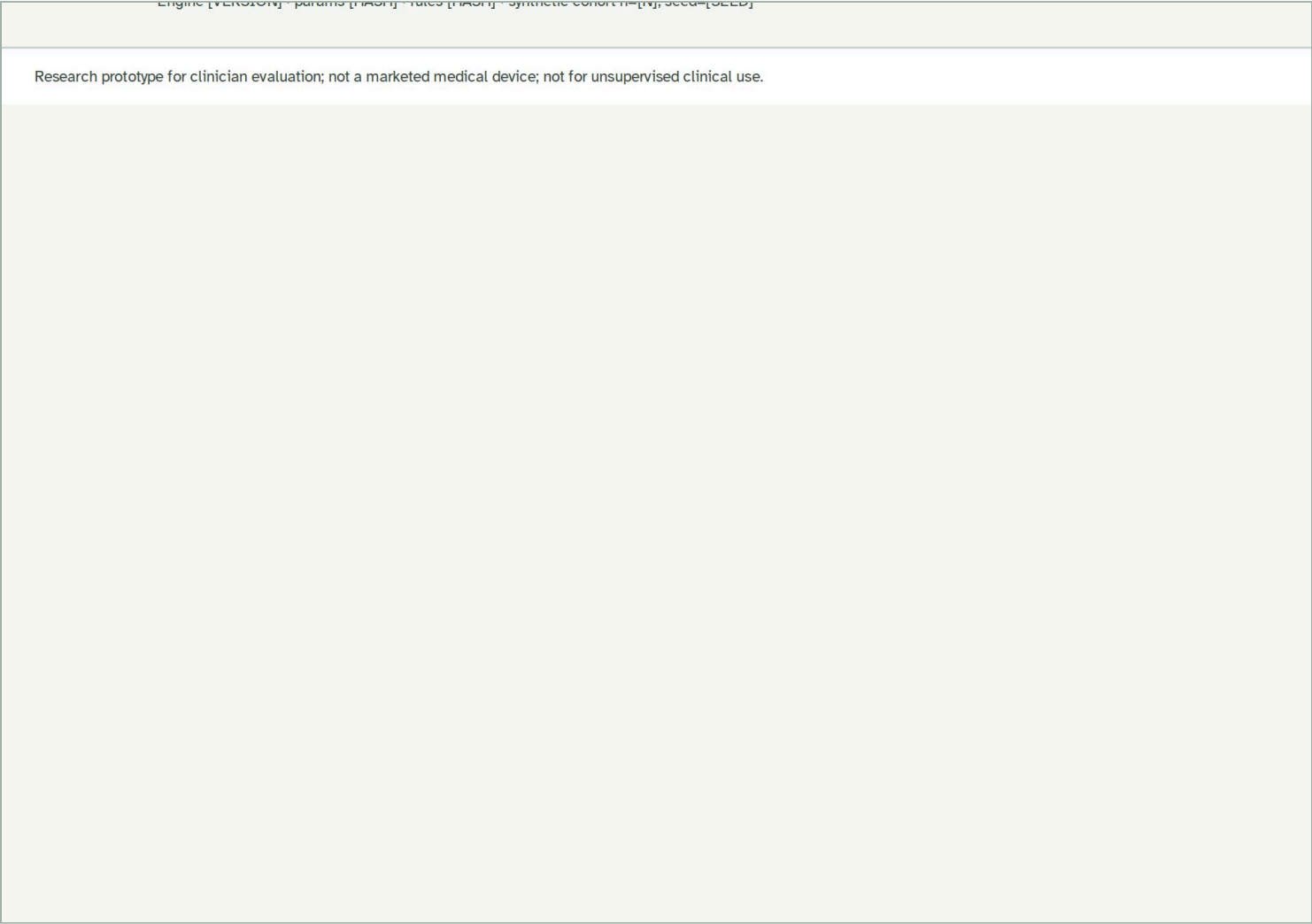
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Guide

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